

Kartik Soni

Member of Technical Staff-1 — Oracle Cloud Infrastructure (OCI)

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B.Tech Computer Science and Engineering — IIT Kanpur

Professional Experience

Oracle Cloud Infrastructure (OCI)

July 2025 – April 2026

Member of Technical Staff-1, Metering Team

- Migrated the **Subscription plugin** from a legacy Loom-based system to the **Oracle Redwood Design System**, enhancing UX for 4 internal dashboards and reducing UI inconsistencies by **35%**.
- Automated **Jenkins** troubleshooting via an **LLM log parser**, cutting **MTTR by 50%** through contextualizing failures against pipeline code and historical build data.
- Architected an **AI-driven workflow** using **MCP** to integrate **Confluence**, **Bitbucket**, and **Jira**, saving **15+ hours weekly** on manual documentation updates.

Software Development Intern — VB Studio

May 2024 – July 2024

- Developed a **Markdown preview** feature for Visual Builder using **Oracle JET**, enabling real-time rendering for **500+ internal users** and improving documentation workflow efficiency by **30%**.
- Designed a **RAG** tool with **ChromaDB** and **LangChain**, evaluating 3 chunking strategies to optimize **LLM response accuracy by 25%** and reduce token consumption by **20%**.

Education

Indian Institute of Technology (IIT) Kanpur

Kanpur, India 2021 – 2025

B.Tech Computer Science and Engineering — CPI: 7.2/10

Nalanda Academy, Kota

Senior Secondary (CBSE XII), 92.8% 2021

AECS No.4, Rawatbhata

Secondary (CBSE X), 96% 2019

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL, Bash, Haskell, Ruby

AI / ML: LangChain, LangGraph, LLM Orchestration, RAG, Machine Learning, scikit-learn, SymPy, Together AI

Frontend: React 19, Streamlit, Oracle JET, Tailwind CSS, shadcn/ui, HTML/CSS

Backend: FastAPI, Express, Node.js, MongoDB, ChromaDB, Pydantic, REST APIs, JWT

DevOps / Cloud: AWS (Lambda, API Gateway, CloudWatch, SAM), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git, OCI

Tools: NumPy, Pandas, Matplotlib, pix2tex, Microsoft Presidio, G-Eval, LaTeX

Key Projects

GuardianHealth — Medical Triage Platform

GitHub | Live

React 19, TypeScript, FastAPI, LangGraph, AWS

- Built a **full-stack medical triage platform** with a **React 19 + TypeScript + Tailwind CSS** frontend and **FastAPI** backend, deployed via serverless **AWS Lambda + API Gateway**.
- Designed an **11-node LangGraph agent pipeline** with 7-layer governance: Medical Firewall, PII Scrubber, ML Classifier, LLM Supervisor, Compliance Agent, and Audit Logger.
- Trained a **Random Forest symptom-disease classifier** on structured clinical data using **scikit-learn** and **pandas**; serialized with **joblib** for production inference.
- Implemented **JWT authentication** with PBKDF2-SHA256 password hashing, **MongoDB** chat persistence, and automated **pytest safety evaluations** ensuring 100% compliance coverage.

JEE Advanced Math Reasoning Agent

GitHub | Live

LangGraph, LangChain, SymPy, Streamlit, Together AI

- Engineered a **7-node LangGraph reasoning pipeline** (analyze → plan → solve → consolidate → verify → reflect → format) that decomposed calculus problems into explicit symbolic solutions via **sandboxed SymPy execution**.
- Implemented an **LLM-driven self-verification loop** with automated error reflection and up to 3 retries, reducing solution error rate from **18% to 4%** across 140+ JEE benchmark problems.
- Integrated local **pix2tex (LatexOCR)** for image-to-LaTeX conversion, enabling handwritten problem input; built a **Streamlit UI** with side-by-side Agent vs. Raw LLM comparison.
- Achieved **7x improvement** over raw LLM baselines on non-frontier models and **36% improvement** on frontier models (**Llama-3.3-70B**) through structured reasoning and **JSON-mode tool-calling**.

Academic Achievements

- Secured **AIR 723 in JEE Mains 2021** (top 0.06%, 99.94th percentile) among 1,000,000+ candidates; achieved **AIR 768 in JEE Advanced 2021**.
- Awarded **KVPY Scholarship** in SA (2020, AIR 322) and SX (2021, AIR 1291) streams; **NTSE Scholar 2019** (NCERT).
- Received **Certificate of Merit** from CBSE for perfect 100/100 in Mathematics (Class X); Merit Position in JSO/JMO 2018.

Leadership & Activities

Secretary, Web Development — Entrepreneurship Cell, IIT Kanpur

July 2022 – March 2023

- Built websites for CA program, ECell Bootcamp, and **ESummit 2022** using **HTML, CSS, JavaScript, React**, serving **15,000+ attendees**; managed a team of **8 developers**.
- Organized the 2.5-day **ESummit 2022** with **15,000+ footfall** and Startup Expo for **50+ companies**.

Secretary — Quiz Club, IIT Kanpur

July 2022 – March 2023

- Organized campus-wide quizzes; coordinated teams for inter-hall General Championship, securing **top 3 positions in 3 of 4 years**.

Relevant Coursework

Data Structures and Algorithms • Software Development and Operations • Computer Organisation • Fundamentals of Computing • Linear Algebra and Differential Equations • Probability and Statistics • Mathematics for Computer Science • OCI Generative AI